StgcDiff: Spatial-Temporal Graph Condition Diffusion for Sign Language Transition Generation

Jiashu He*
hejs@mail.hfut.edu.cn
Hefei University of Technology
Hefei, China

Jiayi He*
hejy4396@mail.hfut.edu.cn
Hefei University of Technology
Hefei, China

Shengeng Tang†
tangsg@hfut.edu.cn
Hefei University of Technology
Hefei, China

Huixia Ben
huixiaben@gmail.com
Anhui University of Science and Technology
Hefei, China

Lechao Cheng
chenglc@hfut.edu.cn
Hefei University of Technology
Hefei, China

Richang Hong
hongrc@hfut.edu.cn
Hefei University of Technology
Hefei, China

Abstract

Sign language transition generation seeks to convert discrete sign language segments into continuous sign videos by synthesizing smooth transitions. However, most existing methods merely concatenate isolated signs, resulting in poor visual coherence and semantic accuracy in the generated videos. Unlike textual languages, sign language is inherently rich in spatial-temporal cues, making it more complex to model. To address this, we propose StgcDiff, a graph-based conditional diffusion framework that generates smooth transitions between discrete signs by capturing the unique spatial-temporal dependencies of sign language. Specifically, we first train an encoder-decoder architecture to learn a structure-aware representation of spatial-temporal skeleton sequences. Next, we optimize a diffusion denoiser conditioned on the representations learned by the pre-trained encoder, which is tasked with predicting transition frames from noise. Additionally, we design the Sign-GCN module as the key component in our framework, which effectively models the spatial-temporal features. Extensive experiments conducted on the PHOENIX14T, USTC-CSL100, and USTC-SLR500 datasets demonstrate the superior performance of our method.

CCS Concepts

• **Computing methodologies** → **Computer vision**; *Natural language processing*; Machine learning.

Keywords

Sign Language Generation, Spatial-Temporal Graph Convolution

*Both authors contributed equally to this research.
†Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
MMGR '25, Dublin, Ireland.
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-2048-2/2025/10
https://doi.org/10.1145/3746266.3762157

ACM Reference Format:
Jiashu He, Jiayi He, Shengeng Tang, Huixia Ben, Lechao Cheng, and Richang Hong. 2025. StgcDiff: Spatial-Temporal Graph Condition Diffusion for Sign Language Transition Generation. In *Proceedings of the 3rd International Workshop on Deep Multimodal Generation and Retrieval (MMGR '25), October 27–28, 2025, Dublin, Ireland.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3746266.3762157

1 Introduction

Sign language is the primary mode of communication for the deaf and hard-of-hearing communities, encompassing rich spatial and temporal information. Although significant advancements have been made in sign language research [24, 45, 46, 54], the field still faces challenges, particularly due to the limited availability of large-scale sentence-level datasets. As illustrated at the top of Figure 1, a potential solution is to synthesize continuous sign language from isolated sign datasets [4, 9, 19, 31, 61]. However, most previous approaches [38, 53] simply splice isolated signs together to form sentences. This method results in unnatural transitions and abrupt motion shifts. Unlike textual language, which primarily relies on the linear arrangement of words, sign language requires smooth transitions between signs to maintain fluency and naturalness while conveying the intended meaning. Without effective transition modeling, directly concatenating individual signs often leads to awkward interruptions or abrupt shifts, making the synthesized sentence difficult to understand. As such, the challenge of seamlessly connecting individual signs to form continuous, meaningful sign language sentences has become a focal point of recent efforts.

Existing works on sign language transition generation [13, 45] primarily focus on achieving smooth transitions, often overlooking the spatial-temporal cues inherent in signs. Specifically, these approaches tend to treat transitions as purely temporal interpolation problems, without explicitly capturing the spatial structure of human joints. As a result, this can lead to overly smooth motion or physically implausible joint movements, especially when there is a significant motion gap between consecutive segments or when complex spatial articulation is involved. Natural transitions between discrete sign language segments rely not only on smooth temporal connections but also on maintaining the continuity of the spatial structure. To ensure the naturalness and physical plausibility

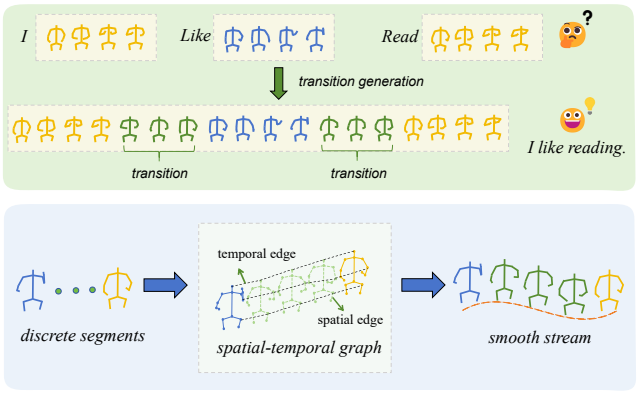


Figure 1: Top: An illustration of the sign language transition task, where transitions (green) are generated between three discrete sign segments (blue and yellow) to produce a continuous sign language video. Bottom: The proposed spatial-temporal graph modeling framework, where temporal and spatial dependencies are jointly leveraged to synthesize smooth transitions between discrete segments.

of transition frames, spatial-temporal modeling becomes crucial. By capturing both the spatial and temporal dependencies in sign language movements, we can better account for the interactions between joints and the continuous flow of motion. This calls for a spatial-temporal representation that accurately conveys both the spatial positions of the joints and their dynamic changes over time, thereby generating more natural and coherent transition frames.

To address this issue, StgcDiff is proposed, a diffusion model designed to generate smooth transitions between discrete sign language segments by effectively capturing the inherent spatial-temporal cues. Specifically, Sign-GCN is introduced, combining a spatial structure extraction module with a temporal feature perception module. A GCN-based encoder-decoder framework is pre-trained, enabling more accurate and detailed reconstructions compared to previous methods. By extracting structure-aware features from the observed frames through the encoder, conditional guidance is provided to the diffusion model, allowing it to generate transitions that preserve the spatial integrity of sign poses while ensuring smooth and coherent temporal motion. The main contributions of this work are summarized as follows:

- A diffusion framework based on spatial-temporal graphs is proposed for sign language transition generation. Additionally, a pre-trained encoder-decoder architecture is employed to learn structure-aware representations of observed sign segments, which serve as conditional information to guide the generation of transition frames.
- We design a Sign-GCN module to effectively model the spatial-temporal dependencies in sign pose sequences. In the spatial domain, a GCN is leveraged to capture the relationships between different joints in the sign language skeleton. In the temporal domain, a multi-scale TCN is employed to capture the long-range temporal dependencies, allowing the module to effectively perceive the evolution of the sign language sequence over time.
- Extensive experiments are conducted on three publicly available sign language datasets, PHOENIX14T[3], USTC-CSL100[20], and USTC-SLR500 [19], demonstrating the superior performance of the proposed method in generating high-quality transitions, surpassing existing approaches.

2 Related Work

2.1 Sign Language Production & Transition

Sign language is the main form of communication for deaf and hard-of-hearing people. Current works focus on building a two-way conversion system from sign language to text. Despite significant progress in Sign Language Translation (SLT) [5, 10, 44, 63] and Recognition (SLR) [29, 64, 65, 67], substantial challenges remain in Sign Language Production (SLP).

Sign language production aims to convert text into real sign language videos or pose sequences. Existing SLP works can be divided into rule-based and data-driven methods. Rule-based methods [1, 7, 25] rely on predefined sign language dictionaries to map text to corresponding signs. However, the construction of sign language dictionaries entails not only substantial costs but also specialized expertise in sign linguistics. With the rapid development of deep learning, data-driven methods have become the mainstream research paradigm in the field of SLP. Saunders *et al.* [39] introduces Progressive Transformers for SLP, which utilize a Transformer-based architecture to enhance the fluency and accuracy of generated sign language. Recently, diffusion-based methods [46, 47, 54] have been applied to SLP, leveraging the unique properties of diffusion models to generate sign pose sequences under semantic guidance.

Despite significant progress, limited long-sequence sign language data remains the key bottleneck for sign language. To address this issue, Zeng *et al.* [59] develops an adaptive interpolation algorithm based on kinematic features that enables natural transition generation. Recently, Tang *et al.* [45] proposes a method converting the unsupervised transition modeling into a supervised learning task via a random masking strategy, which provides a new direction for transitions. However, these methods neglect the inner-frame kinematic dependencies between body joints, causing a lack of coherence and accuracy. To this end, we propose Sign-GCN to generate smooth transitions between discrete symbols by capturing inherent spatial-temporal information.

2.2 Diffusion-based Generative Models

Diffusion models [14, 43] have become a popular generative paradigm due to their ability to produce high-quality and diverse samples by modeling the generation process as the reversal of a gradual noise-adding process. Initially applied in image generation, diffusion models have shown remarkable performance in generating complex structures across multiple domains, including image generation [37], video synthesis [30, 56], 3D shape modeling [50], and human motion generation [62, 66].

Despite their success in vision and motion tasks, the application of diffusion models in Sign Language Production (SLP) remains relatively underexplored. Recent efforts such as G2P-DDM [54] introduce discrete diffusion techniques to map gloss sequences to pose sequences, while VQ-GCDM [47] employs a gloss-conditioned multi-hypothesis diffusion framework for diverse sign generation.

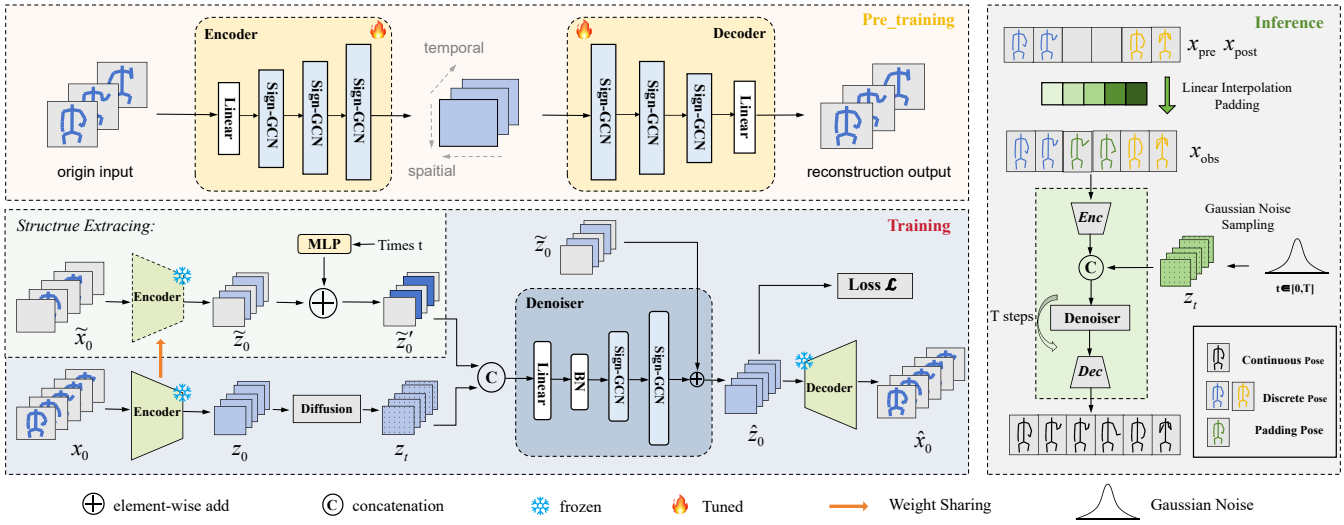


Figure 2: The overview of the proposed method. In the pre-training phase (left top), a long-duration sign video is processed by our gragh-based encoder-decoder architecture to reconstruct the input, which aims to learn the spatial-temporal representation. In the training phase (left bottom), we train a conditional diffusion model to generate smooth transitions between discrete segments. We sample the noisy sign skeleton features z_t following the diffusion process. The diffusion denoiser predicts the feature of transitions from z_t guided by the learned representation extracted from the randomly masked input video $\tilde{x}_0$. For optimization, MAE loss $\mathcal{L}_{MAE}$ is utilized in the training stage. In the inference phase (right), we apply the linear interpolation padding to initialize transition frames aiming to connect discrete segments. Then, our diffusion model (green part) refines these transitions by iteratively denoising T steps from gaussian noise, generating a continuous sign video coherent with both context segments.

However, these approaches primarily focus on the semantic correctness of generated poses from linguistic inputs and overlook the challenge of maintaining temporal continuity, especially across discrete sign segments. To this end, Tang *et al.* [45] proposes Sign-D2C, a conditional diffusion model to synthesize contextually smooth transition frames, enabling the seamless construction of continuous sign language sequences. Our work complements these studies by leveraging a spatial-temporal graph conditional diffusion framework for generating smooth transition frames.

2.3 Graph Convolutional Networks

Graph Convolutional Networks (GCNs) [28] have been widely applied in many applications like visual recognition [15–18, 34], object detection [22, 52], action localization [23, 60], trajectory prediction [41], and image captioning [8]. Many advances [6, 33, 42, 48, 57] have demonstrated the potential of GCNs in modeling the complex spatial-temporal patterns inherent in skeletons. Among these methods, Yan *et al.* [55] proposes Spatial-Temporal Graph Convolutional Networks (ST-GCNs), which first demonstrate the advantages of GCNs in spatial-temporal modeling for human action recognition.

Motivated by these advances in action recognition, GCNs have also been widely adopted in sign language, such as Sign Language Recognition (SLR) [2, 12, 24, 26, 49, 51]. For instance, Jiang *et al.* [24] introduces SL-GCN and SSTCN within a multi-modal framework to enhance isolated SLR accuracy. Tunga *et al.* [49] proposes a model that combines GCN and BERT to effectively capture spatial and temporal features from skeleton sequences for pose-based sign language recognition. Recent studies such as Parelli *et al.* [36] and Arib *et al.* [2] propose incorporating GCNs into continuous recognition and translation pipelines. Compared to traditional methods, GCNs offer superior representational power for feature extraction and modeling of sign language skeleton sequences. While previous works have leveraged GCNs primarily for recognition and translation tasks, our method focuses on sign language generation, specifically addressing smooth spatial-temporal transitions between discrete sign segments, a relatively unexplored aspect in GCN-based SLP [58].

Inspired by Spatial-Temporal Graph Convolutional networks [55], we propose a graph-based sign language transition. Our method eliminates the previous neglect of local skeletal structural consistency and models sign poses in a spatial-temporal skeleton graph with high granularity.

3 Preliminaries: Unsupervised to Supervised

Generating smooth transitions between discrete sign segments is challenging due to the lack of ground-truth transition data, as real-world sign datasets often lack explicitly annotated transitions.

To address this, we follow the core idea proposed in Sign-D2C [45] and transform the problem into a supervised learning task through a random masking strategy. Specifically, during training, long continuous sign language sequences are treated as complete ground-truth trajectories. We randomly mask out short temporal segments within these sequences, effectively simulating missing transitions

between adjacent observed segments. This masked data setup creates pseudo-gaps similar to those found when stitching together isolated signs in practice.

Given a complete sign language sequence $X = \{x_1, x_2, \ldots, x_T\}$ with T frames, and each x_t represents the 3D coordinates of N joints at time t. We define a masking set $\mathcal{M} \subset \{1, 2, \ldots, T\}$ containing randomly selected time indices. The observed frames are:

$$X_{obs} = \{x_t \mid t \notin \mathcal{M}\}, \quad X_{mask} = \{x_t \mid t \in \mathcal{M}\} \tag{1}$$

The model is then trained to predict the masked frames X_{mask} based on the surrounding observable context X_{obs}, thereby learning to reconstruct smooth, temporally consistent transitions in a supervised manner.

4 Methodology

In this work, we propose a graph-based framework for generating smooth transitions between discrete sign segments to create continuous sign language videos. This section will outline the sign pose modeling (Sec. 4.1), followed by the perceptual structure representation (Sec. 4.2), the overall framework (Sec. 4.3) and the Sign-GCN module (Sec. 4.4).

4.1 Spatial-Temporal Modeling of Sign Pose

Let $X \in \mathbb{R}^{C\times T\times V}$ denote the input sign pose sequence, where C is the input channel ($C = 3$ for 3D coordinates), T is the number of frames, and V is the number of joints. To model both spatial and temporal dependencies for the sign language transition task, we construct a spatial-temporal graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v_i^t \in \mathcal{V}$ corresponds to joint i at time t, and the edge set $\mathcal{E}$ is defined as:

$$\mathcal{E} = \underbrace{\{(v_i^t, v_j^t)\}}_{\text{Spatial edges}} \cup \underbrace{\{(v_i^t, v_i^{t+1})\}}_{\text{Temporal edges}}, \tag{2}$$

where the spatial edge (v_i^t, v_j^t) denotes joints i and j are physically connected at time t, and the temporal edge (v_i^t, v_i^{t+1}) denotes the same joint i across consecutive time steps t and $t + 1$.

The spatial edges model the natural anatomical structure of the sign pose, while temporal edges connect the same joint across consecutive frames to capture motion dynamics. By explicitly modeling both spatial and temporal dependencies, the spatial-temporal graph provides a strong structural prior for the diffusion model.

4.2 Structure-aware Representation Learning

In this section, we introduce a GCN-based encoder-decoder pre-training architecture and pre-train it prior to training the full diffusion framework, which aims to comprehensively capture sign poses and learn a structure-aware representation of spatial-temporal skeleton sequences.

As shown in Figure 2, our encoder-decoder architecture contains an encoder and decoder, and we adopt Sign-GCN as the backbone of the encoder-decoder architecture. The encoder $\mathcal{E}$ consists of three stacked Sign-GCN layers that map sign poses to the latent space with rich spatial-temporal information. The decoder $\mathcal{D}$ mirrors the structure of the encoder with three Sign-GCN layers, which reconstruct the hidden embedding under the supervision of the reconstruction criterion.

Given a sign pose sequence $\mathcal{X} \in \mathbb{R}^{3\times T\times V}$ as the input of our encoder-decoder architecture, we firstly embed all skeleton joints and their topology into the spatial-temporal graph $\mathcal{G}$. Then, $\mathcal{X} \in R^{3\times T\times V}$ is linearly transformed to $\mathcal{X}' \in R^{C\times T\times V}$ with learnable parameters. In practice, we set C to 8. With the Encoder $\mathcal{E}$ based on Sign-GCN, we gradually project the input into higher-dimensional spaces with dimensions 16, 64, and 128, resulting in a latent representation $Z \in \mathbb{R}^{128\times T\times V}$. The decoder progressively reduces the latent feature dimension from 128 to 64, then to 16, and finally reconstructs the input sequence back into the original pose domain as $\hat{\mathcal{X}} \in \mathbb{R}^{3\times T\times V}$ with a trainable linear layer. Moreover, we employ batch normalization for $\mathcal{X}'$ in the encoder. The process of our reconstruction is formalized as:

$$\begin{cases} Z = \mathcal{E}(\mathcal{X}), & Z \in \mathbb{R}^{128\times T\times V}, \\ \hat{\mathcal{X}} = \mathcal{D}(Z), & \hat{\mathcal{X}} \in \mathbb{R}^{3\times T\times V} \end{cases} \tag{3}$$

The objective of pre-training can be formalized as minimizing the divergence between $\mathcal{X}$ and $\hat{\mathcal{X}}$. In this paper, we optimize the encoder-decoder architecture using the Mean Absolute Error (MAE) loss to constrain the consistency of the reconstructed output $\hat{\mathcal{X}}$ and the input pose sequence $\mathcal{X}$.

4.3 StgcDiff Framework

In this section, we describe the proposed conditional diffusion framework StgcDiff, as a unified framework for sign language transition. Figure 2 illustrates the overall pipeline of our method.

Conditioning. Given the sign video dataset p_{video}, we take a T-frame original video denoted as $x_0 \in \mathbb{R}^{3\times T\times V}$, where x_0 follows the distribution p_{video}. Following [45], we randomly mask out the original video x_0 to simulate the absence of transition frames, resulting in a masked sign video $\tilde{x}_0 \in \mathbb{R}^{3\times T\times V}$. Then, $\tilde{x}_0$ is encoded by the pretrained GCN-encoder $\mathcal{E}$ as $\tilde{z}_0 \in \mathbb{R}^{C\times T\times V}$, where C, T and V indicate the channels, the frame number and the joint number. Meanwhile, we employ a MLP for the times of adding noise t and the time embedding is added to $\tilde{z}_0$ to incorporate temporal information:

$$\tilde{z}_0' = \tilde{z}_0 + \mathrm{MLP}(t), \tag{4}$$

where $\tilde{z}_0'$ denotes the conditional guidance for the generation of sign language transition frames.

Denoiser Architecture. Our denoising module takes as input the noised transition frames along with context information and time embedding, which reconstructs the original representation by iteratively removing the noise ϵ. First, we apply linear transformations to project the input into a new feature space of $\mathbb{R}^{F'\times T\times V}$. Then, we put the feature map to two Sign-GCNs, which is responsible for extracting features in dimensions 64 and 128 respectively. To do this, we can obtain a new feature. In additional to further ensure the generated transitions are guided by context information, we also inject the masked encoder output $\tilde{z}_0$ into the final residual connection, which helps preserve semantic relevance and stabilizes the denoising process.

Training Phase. In the training stage, the latent code z_0 of sign video is corrupted by Gaussian noise ϵ as z_t whose size is same as $z_0 \in \mathbb{R}^{C\times T\times V}$. This process is formulated as follows:

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{5}$$

In order to incorporate with sign observations which provide the necessary context for our model to predict the masked segments accurately, our denoiser takes z_t concatenated with the condition $\tilde{z}_0'$ as the input:

$$\tilde{z_t} = [z_t, \tilde{z}_0'], \quad \tilde{z}_t \in \mathbb{R}^{(2\times C)\times T\times V} \tag{6}$$

where $\tilde{z}_t$ represents the final input to the denoiser. Finally, we utilize the denoiser to gain $\hat{z}_0$, the final prediction of the same shape as z_0.

During the diffusion model training phase, we employ the denoising loss $\mathcal{L}_{denoise}$ to ensure that the denoiser's predictions $\hat{z}_0$ remain consistent with the original latent features z_0 :

$$\mathcal{L}_{denoise} = \mathbb{E}_{z_0,t}\left[|z_0 - \hat{z}_0|\right] \tag{7}$$

The denoiser is trained by minimizing the loss $\mathcal{L}_{denoise}$ directly in an end-to-end manner.

Inference Phase. At inference time, our goal is to generate smooth intermediate frames between discrete sign segments, resulting in seamless synthesis of continuous sign language videos. The transition segment $x_{\text{trans}} \in \mathbb{R}^{3\times T_{trans}\times V}$ is initialized using a linear interpolation between the last frame of the previous segment $x_{\text{pre}} \in \mathbb{R}^{3\times T_{pre}\times V}$ and the first frame of the next segment $x_{\text{post}} \in \mathbb{R}^{3\times T_{post}\times V}$, where T_{trans}, T_{pre} and T_{post} respectively represent the number of transition frames, previous observation frames and next observation frames. We aggregate the three segments to obtain a entire video sequence as x_{obs}. The above description can be expressed as:

$$x_{obs} = Concat(x_{pre}, x_{trans}, x_{post}) \tag{8}$$

Then, the encoder extracts a latent representation from x_{obs}, formulated as $z_{obs} = \mathcal{E}(x_{obs})$, which provides guidance for the process of denoising. Starting from pure Gaussian noise z_t, the model progressively denoises the sequence across t steps conditioned on z_{obs}:

$$\hat{z}_{t-1} = \text{Denoiser}(z_t, t, z_{\text{obs}}) \tag{9}$$

until the final output $\hat{z}_0$ is obtained. Finally, the decoder maps the final latent representation $\hat{z}_0$ back to the pose space by pretrained decoder $\mathcal{D}$, producing smooth and natural transitions that are coherent with both context segments.

4.4 Sign-GCN Module

As shown in Figure 3, we design a module named Sign-GCN that integrates spatial and temporal modeling in a unified structure. This module is composed of three main parts: a Spatial Separation Layer for adjacency matrix construction, a spatial graph convolution layer for spatial modeling, and a multi-scale temporal convolutional network for temporal modeling.

Firstly, the adjacency matrices are constructed and normalized based on spatial strategies (*i.e.*, center, centripetal, and centrifugal) introduced in [55], resulting in three spatial adjacency matrices $A_k k = 1^3 \in \mathbb{R}^{3\times V\times V}$. Next, the feature map $fin \in \mathbb{R}^{C_{\text{in}}\times V\times T}$ is input to the spatial graph convolutional layer, where C_{in} represents the number of input channels, V is the number of nodes, and T is the time dimension. Following the Graph Convolution formulation proposed by Kipf and Welling [28], the output of the graph convolutional layer is computed as:

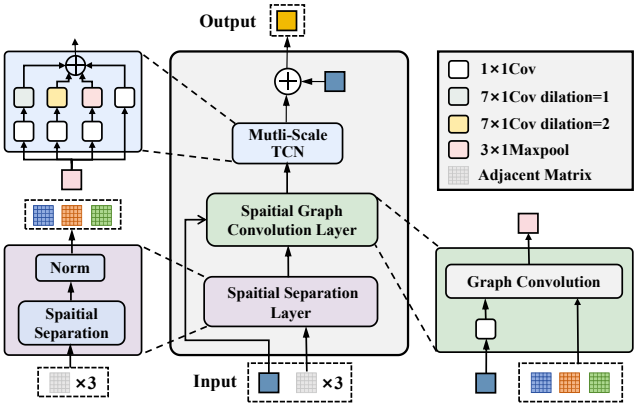


Figure 3: The main components of the Sign-GCN module, include the Spatial Separation Layer, Spatial Graph Convolution Layer, and Multi-Scale Temporal Convolutional Network (TCN), which together model the spatial and temporal dependencies in sign language sequences.

$$f_{\text{out}} = \sum_{k}^{k_s} (f_{\text{in}} \cdot A_k) W_k, \tag{10}$$

$$A_k = D_k^{-\frac{1}{2}}(\tilde{A}_k + I)D_k^{-\frac{1}{2}}, \quad D^{ii} = \sum_j (\tilde{A}_k^{ij} + I_{ij}), \tag{11}$$

where $f_{\text{out}} \in \mathbb{R}^{C_{\text{out}}\times V\times T}$ is the output feature map of the spatial graph convolution layer, k_s denotes the number of spatial kernels (k_s=3), $\tilde{A}_k$ is the undirected adjacency matrix representing intra-body connections, I is the identity matrix, and W_k is the trainable weight matrix corresponding to each adjacency matrix.

In the temporal domain, we introduce a Multi-scale TCN to capture temporal features in consecutive frames. As shown in Figure 3, the Multi-TCN module contains four parallel branches, each containing a 1×1 convolution to reduce channel dimension to $C_{out}/4$. Two branches contain two temporal convolutions with different dilation factors d , each of which uses a $k_t \times 1$ $2D$ convolution operating on (T, V), where k_t is the temporal kernel size. And one branch contains a 3×1 MaxPool layer following 1×1 convolution. Finally, the outputs from all four branches are concatenated along the channel dimension and summed with the residual connection to obtain the final output y.

With the spatial graph convolution layer and the Multi-scale TCN, our Sign-GCN network can extract and accumulate spatial and temporal interactive features simultaneously.

5 Experiments

5.1 Experimental Settings

Datasets. We evaluate our method on three publicly available sign language datasets: PHOENIX14T [3], USTC-CSL100 [20], and USTC-SLR500 [19]. The PHOENIX14T dataset comprises 8,257 videos spanning 2,887 distinct German words and 1,066 glosses, renowned for its inherent complexity. The USTC-CSL100 dataset contains 100 Chinese sign language sentences performed by 50 signers, which

Table 1: Ablation results of modules on PHOENIX14T.

Methods	DEV			TEST		
	B1↑	B4↑	ROUGE↑	B1↑	B4↑	ROUGE↑
Base	19.32	5.92	18.33	19.59	6.26	18.59
Base+E-D	21.12	7.16	21.71	19.97	6.54	20.01
Base+E-D+Denoiser	**21.55**	**7.33**	**21.87**	**20.69**	**6.98**	**20.47**

Table 2: Ablation results of parameters on PHOENIX14T.

Model Config	Module	DEV			TEST		
		B1↑	B4↑	DTW↓	B1↑	B4↑	DTW↓
7×1 Convolution	TCN	20.90	7.20	0.45	20.24	6.77	0.45
Multi-scale (Ours)	TCN	**21.55**	**7.33**	**0.32**	**20.69**	**6.98**	**0.32**
w/o residual	Denoiser	20.73	6.91	0.49	19.97	6.54	0.49
w/ residual (Ours)	Denoiser	**21.55**	**7.33**	**0.32**	**20.69**	**6.98**	**0.32**

Table 3: Ablation results of parameters on PHOENIX14T.

Param	Value	DEV			TEST		
		B1↑	B4↑	ROUGE↑	B1↑	B4↑	ROUGE↑
k	3	20.98	7.18	21.52	19.94	6.70	19.98
	5	20.93	7.12	21.71	20.52	6.82	**20.88**
	7	**21.55**	**7.33**	**21.87**	**20.69**	**6.98**	20.47
	9	20.67	6.93	21.47	20.24	6.62	20.45
d	1	20.75	7.05	21.51	20.12	6.73	20.18
	2	**21.55**	**7.33**	**21.87**	**20.69**	**6.98**	**20.47**
	3	20.81	6.93	21.56	20.00	6.76	20.03
	4	21.16	7.26	21.76	20.03	6.64	20.02
r	0.1	21.15	7.19	21.67	19.75	6.67	19.66
	0.4	21.19	7.14	21.66	19.77	6.66	19.68
	0.5	**21.55**	**7.33**	**21.87**	**20.69**	**6.98**	**20.47**
	0.6	21.14	7.10	21.52	20.20	6.90	20.08

we partitioned into 4,000 training instances and 1,000 test instances following specifications in [11] . The USTC-SLR500 contains 500 isolated Chinese signs performed by 50 signers. Following Huang *et al.* [19], we divided the 36 and 14 signers into training sets and test sets.

Implementation Details. The encoder and decoder each consist of 3 Sign-GCN layers, while the denoiser has 2 Sign-GCN layers. In the encoder, the input-output channels of Sign-GCNs are 8-16-64-128. In the denoiser, the channels are 32-64-128. The decoder is symmetric to the encoder, with the dimensional transformations in reverse order. In addition, we set the timesteps t of the diffusion model to 1,000 and the number of inferences i to 5. During training, we use the Adam optimizer [27] and a learning rate of 1×10^{-3}. Experiments are conducted using PyTorch on NVIDIA GeForce RTX 2080 Ti GPUs.

Evaluation Metrics. Due to the current lack of datasets for sign language transition, following prior works [45], we simulate realistic discrete sign segments by implementing an interval sampling strategy (remove Y frames every X frames) on large-scale continuous sentence datasets (*i.e.*, PHOENIX14T, USTC-CSL100). We observe that the duration of commenting on sign language is nearly 20 frames. So in our experiments, we mainly adopt the method of removing 20 frames every 30 frames (Remove 20 frames - Every 30 frames) and the method of removing 10 frames every 30 frames (Remove 10 frames - Every 30 frames) to evaluate our performance. Furthermore, we use the isolated sign language dataset (*i.e.*, USTC-SLR500) to assemble discrete vocabulary videos into continuous sequences, demonstrating the model's capability to produce smooth transitions in a real-world setting. We assess the generated transitions from two aspects: ***semantic accuracy*** and ***temporal coherence***. To evaluate the semantic accuracy of generated frames, we use NSLT [3] to convert generated sign language videos back into text, following prior works [21, 39, 40] in SLP. Then we compare this text with reference data to calculate semantic accuracy metrics including BLEU [35], ROUGE [32], and WER, while we use DTW to evaluate the temporal coherence of results.

5.2 Ablation Study

In this subsection, we explore the effectiveness of Sign-GCN, Multi-scale TCN and related parameters. The experiments are conducted under the condition of removing 20 frames out of every 30 frames (Remove 20 frames - Every 30 frames) to evaluate the semantic accuracy and motion coherence.

Effectiveness of Sign-GCN. Table 1 shows the ablation results of Sign-GCN module. We use Sign-D2C [45] as the baseline model, denoted as Base. Base+E-D refers to using Sign-GCN instead of the encoder-decoder architecture in Base, which increases BLEU-1/BLEU-4 by 0.38%/0.28% on the TEST set, respectively. When Sign-GCN is used to replace both Encoder-Decoder and Denoiser, the performance is further improved, with BLEU-4 reaching 7.33% and 6.98% on the DEV set and TEST set, respectively . This indicates that Sign-GCN preserves richer and more accurate semantics.

Effectiveness of Multi-scale Temporal Modeling. To evaluate the impact of multi-scale temporal feature extraction, we compare our proposed Multi-scale TCN with a standard 7×1 convolution-based TCN under a controlled setting. As shown in Table 2, our Multi-scale TCN yields significant improvements across all metrics on both the DEV and TEST sets. Specifically, on the TEST set, Multi-scale TCN surpasses the baseline by +0.45% BLEU-1, +0.21% BLEU-4 and -0.13% DTW. This highlights Multi-scale TCN's ability to pick up complex temporal correlations that promotes smooth and accurate flow in generated sign videos.

Residual Connections in Denoiser. We further investigate the role of residual connections within the denoising module as shown in Table 2 . Removing residual connections (w/o residual) results in a noticeable performance drop, with the TEST DTW score increasing from 0.32% to 0.49%. Similarly, both BLEU-1 and BLEU-4 metrics decline. The residual-enhanced variant (w/ residual) achieves the improvement of overall performance. These results strongly demonstrate that residual connections of condition information can preserve more semantic relevance for our model.

Exploration of parameters. Table 3 compares various Sign-GCN settings, including different values of k, d, where k represents the temporal kernel size, and d represents the dilation rate for the

Table 4: Performance comparison on PHOENIX14T (under setting: Remove 10 frames - Every 30 frames).

Methods	DEV					TEST				
	B1↑	B4↑	ROUGE↑	WER↓	DTW↓	B1↑	B4↑	ROUGE↑	WER↓	DTW↓
Ground Truth	29.77	12.13	29.60	74.17	0.00	29.76	11.93	28.98	71.94	0.00
G2P-DDM [54]	8.25	0.99	7.17	98.93	12.71	9.52	1.29	8.12	98.54	12.57
VQ-GCDM [47]	9.31	0.87	7.59	98.69	12.11	9.85	0.96	7.95	98.59	11.98
Sign-D2C [45]	22.46	8.06	22.07	80.73	0.62	22.36	**8.73**	22.36	79.06	0.61
Ours	**23.74**	**8.51**	**24.15**	**77.69**	**0.32**	**22.64**	8.48	**23.08**	**76.59**	**0.32**

Table 5: Performance comparison on PHOENIX14T (under setting: Remove 20 frames - Every 30 frames).

Methods	DEV					TEST				
	B1↑	B4↑	ROUGE↑	WER↓	DTW↓	B1↑	B4↑	ROUGE↑	WER↓	DTW↓
Ground Truth	29.77	12.13	29.60	74.17	0.00	29.76	11.93	28.98	71.94	0.00
G2P-DDM [54]	8.98	0.63	7.44	98.61	15.95	9.13	0.74	7.75	98.64	15.58
VQ-GCDM [47]	8.64	0.89	7.08	98.48	12.40	9.02	1.36	7.22	98.61	12.22
Sign-D2C [45]	19.32	5.92	18.33	87.30	0.70	19.59	6.26	18.59	86.80	0.68
Ours	**21.55**	**7.33**	**21.87**	**85.21**	**0.32**	**20.69**	**6.98**	**20.47**	**84.50**	**0.32**

Table 6: Performance comparison on USTC-CSL100 (under setting: Remove 20 frames - Every 30 frames).

Methods	B1↑	B4↑	ROUGE↑	WER↓	DTW↓
Ground Truth	69.10	59.12	68.53	47.38	0.00
G2P-DDM [54]	13.43	1.74	13.62	92.15	7.47
VQ-GCDM [47]	18.77	2.08	16.64	92.67	10.89
Sign-G2C [45]	67.95	56.61	**67.38**	**49.24**	1.79
Ours	**68.07**	**57.11**	67.15	49.38	**1.40**

multi-scale TCN operations. We first observe that all configurations consistently outperform the baseline, confirming the stability of Sign-GCN as a spatial-temporal feature extractor. Specifically, increasing the kernel size k from 3 to 7 leads to a consistent improvement across all metrics, with the best performance observed at k=7, achieving BLEU-1 scores of 21.55% on DEV and 20.69% on TEST. However, further increasing k to 9 results in a slight drop in performance. As for the dilation rate d, we find that the performance peaks at d=2, with ROUGE scores of 21.87% on DEV and 20.47% on TEST. Both lower dilation and higher dilation lead to marginal performance drops. This suggests that moderate dilation enables a better balance between capturing local dependencies and maintaining global contextual awareness. These results confirm that k=7 and d=2 is an effective configuration for Sign-GCN. Meanwhile, table 3 also shows masking ratio r = 0.5 is a good balance between semantic retention and transition smoothness.

5.3 Comparison with State-of-the-Arts

Due to the current lack of benchmarks for sign language transition generation, we adapted the SLP models (*i.e.* G2P-DDM [54] and VQ-GCDM [47]) as benchmarks. Additionally, we compare with the sign language transition method (*i.e.* Sign-D2C [45]).

PHOENIX14T Table 4 shows the results under the "Remove 10 frames - Every 30 frames" setting, StgcDiff significantly outperforms two compared models G2P-DDM and VQ-GCDM, achieving 23.74% and 22.64% BLEU-1 on the DEV and TEST sets, respectively. Even compared with the best-performing method Sign-D2C, our method achieves better performance on most metrics. Furthermore, our method shows significant gains in temporal coherence, *e.g.*, StgcDiff is 0.30% and 0.29% lower DTW than Sign-D2C [45] on DEV and TEST. These results confirm that our model generates smoother and more realistic transitions than the baseline methods.

To further prove the performance of our method, we compare with other methods under the setting "Remove 20 frames - Every 30 frames" as shown in Table 5. In this challenging setting, our method achieves superior performance on the all metrics. We achieve BLEU-1/BLEU-4 scores of 21.55%/7.33% on DEV and 20.69%/6.98% on TEST. Meanwhile, our method obtains higher ROUGE and lower WER scores, demonstrating the better performance in semantic relevance. Moreover, our method is especially significantly better than Sign-D2C by DTW score of 0.32 on the DEV and TEST sets, further highlighting the advantages of our approach in accurately generating natural transition poses.

USTC-CSL100 Table 6 shows comparison results on a challenging Chinese sign language benchmark USTC-CSL under the more challenging setting "Remove 20 frames - Every 30 frames". Our proposed method achieves competitive performance, with a BLEU-1



Figure 4: Visualization examples of generating transition pose under setting "Remove 10 frames - Every 30 frames" on PHOENIX14T. We compare our method with three SOTA methods, attached the Ground Truth. Green: Transitions. Orange and Blue: Observations.

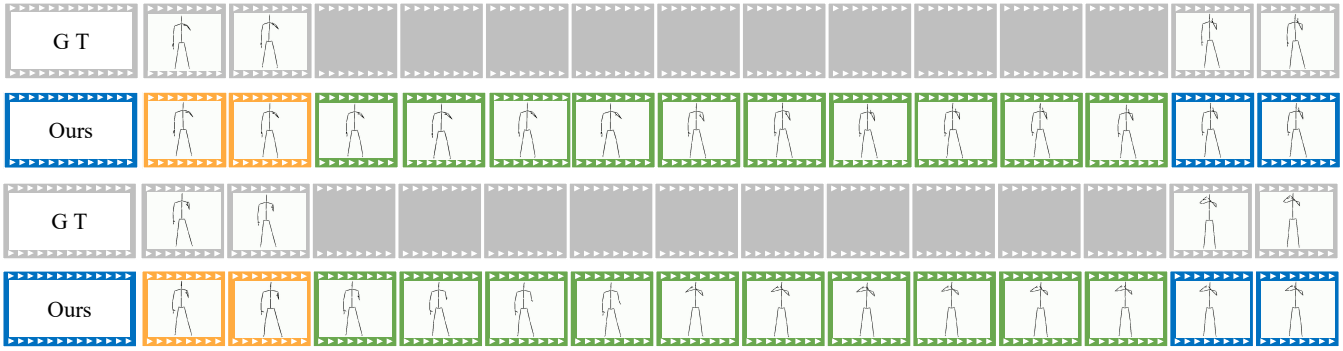


Figure 5: Visualization examples on the USTC-SLR500, demonstrate the generation of transition poses in a realistic word-level scenario. Green: Transitions.Orange and Blue: Observations.

score of 68.07% and a BLEU-4 score of 57.11%, which are higher than other baseline methods. In terms of ROUGE, our method performs similarly to the best methods, reaching 67.15%. For WER, our model also shows a notable result with a score of 49.38%, which is close to the performance of the best-performing method Sign-D2C. Regarding DTW, our method shows the best results with values of 1.40, indicating better alignment and more coherent transitions compared to other methods.

5.4 Qualitative Results

Visualization Results. In Figure 4, we visualize transition sign poses generated by the proposed StgcDiff and other methods, *i.e.*, G2P-DDM [54], VQ-GCDM [47] and Sign-D2C [45] . This example clearly shows the transition frames generated by StgcDiff are noticeably superior to another three methods. Specifically, our method significantly outperforms G2P-DDM and VQ-GCDM in terms of temporal coherence, as the latter two exhibit unnatural abruption. In comparison of Sign-D2C, our method has an advantage in spatial modeling of skeleton joints, further highlighting the good spatial representation learning ability of our proposed model.

Verification in Realistic Scenarios. Figure 5 presents two visual examples illustrating the synthesis of continuous sign language videos from discrete segments in a realistic word-level context, based on the USTC-SLR500 dataset. In each example, the observed frames from the two separate sign segments are highlighted in blue and orange, while the generated transition frames are depicted in green. The visualizations indicate that our approach produces transitions that are not only temporally smooth but also spatially consistent, effectively bridging the gap between the isolated segments and yielding a seamless, coherent sign language video.

6 Conclusions

In this work, we propose a novel graph-based diffusion framework for sign transition generation, effectively capturing spatial-temporal dependencies to produce smooth and contextually accurate transitions. The designed Sign-GCN module models both the spatial relationships between joints and the multi-scale temporal dependencies in sign language sequences. Extensive experiments on three benchmarks validate the superior performance of the proposed method.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grants No. 62472139, U23B2031), the Anhui Provincial Natural Science Foundation, China (Grant No. 2408085QF191), the Fundamental Research Funds for the Central Universities (Grants No. PA2025IISL0112, JZ2024HGTA0178), and the Major Project of Anhui Province (Grant No. 202423k09020001).

References

[1] Ahmed H Aliwy and A Alethary Ahmed. 2021. Development of Arabic Sign Language Dictionary using 3D Avatar Technologies. *Indonesian Journal of Electrical Engineering and Computer Science* (2021), 609–616.

[2] Safaeid Hossain Arib, Rabeya Akter, Sejuti Rahman, and Shafin Rahman. 2025. SignFormer-GCN: Continuous Sign Language Translation using Spatio-temporal Graph Convolutional Networks. *PloS one* (2025), e0316298.

[3] Necati Cihan Camgoz, Simon Hadfield, Oscar Koller, Hermann Ney, and Richard Bowden. 2018. Neural Sign Language Translation. In *IEEE Conference on Computer Vision and Pattern Recognition*. 7784–7793.

[4] Xiujuan Chai, Hanjie Wang, and Xilin Chen. 2014. The DEVISIGN Large Vocabulary of Chinese Sign Language Database and Baseline Evaluations. In *Technical report, Key Lab of Intelligent Information Processing of CAS*. Institute of Computing Technology.

[5] Yutong Chen, Fangyun Wei, Xiao Sun, Zhirong Wu, and Stephen Lin. 2022. A Simple Multi-modality Transfer Learning Baseline for Sign Language Translation. In *IEEE Conference on Computer Vision and Pattern Recognition*. 5120–5130.

[6] Yuxin Chen, Ziqi Zhang, Chunfeng Yuan, Bing Li, Ying Deng, and Weiming Hu. 2021. Channel-Wise Topology Refinement Graph Convolution for Skeleton-Based Action Recognition. In *International Conference on Computer Vision*. 13359–13368.

[7] Mathieu De Coster and Joni Dambre. 2023. Querying a Sign Language Dictionary with Videos using Dense Vector Search. In *IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops*. 1–5.

[8] Xinzhi Dong, Chengjiang Long, Wenju Xu, and Chunxia Xiao. 2021. Dual Graph Convolutional Networks with Transformer and Curriculum Learning for Image Captioning. In *ACM International Conference on Multimedia*. 2615–2624.

[9] Philippe Dreuw, Carol Neidle, Vassilis Athitsos, Stan Sclaroff, and Hermann Ney. 2008. Benchmark Databases for Video-Based Automatic Sign Language Recognition. In *International Conference on Language Resources and Evaluation*.

[10] Jia Gong, Lin Geng Foo, Yixuan He, Hossein Rahmani, and Jun Liu. 2024. Llms Are Good Sign Language Translators. In *IEEE Conference on Computer Vision and Pattern Recognition*. 18362–18372.

[11] Dan Guo, Wengang Zhou, Houqiang Li, and Meng Wang. 2018. Hierarchical LSTM for Sign Language Translation. In *AAAI Conference on Artificial Intelligence*. 6845–6852.

[12] Qi Guo, Shujun Zhang, Liwei Tan, Ke Fang, and Yinghao Du. 2023. Interactive Attention and Improved GCN for Continuous Sign Language Recognition. *Biomedical Signal Processing and Control* (2023), 104931.

[13] Jiayi He, Xu Wang, Shengeng Tang, Yaxiong Wang, Lechao Cheng, and Dan Guo. 2025. Motion is the Choreographer: Learning Latent Pose Dynamics for Seamless Sign Language Generation. arXiv:2508.04049 [cs.CV] https://arxiv.org/abs/2508.04049

[14] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. *Neural Information Processing Systems* (2020), 6840–6851.

[15] Danfeng Hong, Lianru Gao, Jing Yao, Bing Zhang, Antonio Plaza, and Jocelyn Chanussot. 2020. Graph Convolutional Networks for Hyperspectral Image Classification. *IEEE Transactions on Geoscience and Remote Sensing* (2020), 5966–5978.

[16] Tao Hu, Chengjiang Long, and Chunxia Xiao. 2021. A Novel Visual Representation on Text using Diverse Conditional GAN for Visual Recognition. *IEEE Transactions on Image Processing* (2021), 3499–3512.

[17] Tao Hu, Chengjiang Long, and Chunxia Xiao. 2024. CRD-CGAN: Category-Consistent and Relativistic Constraints for Diverse Text-to-Image Generation. *Frontiers of Computer Science* (2024), 181304.

[18] Gang Hua, Chengjiang Long, Ming Yang, and Yan Gao. 2013. Collaborative Active Learning of a Kernel Machine Ensemble for Recognition. In *International Conference on Computer Vision*. 1209–1216.

[19] Jie Huang, Wengang Zhou, Houqiang Li, and Weiping Li. 2018. Attention-Based 3D-CNNs for Large-Vocabulary Sign Language Recognition. *IEEE Transactions on Circuits and Systems for Video Technology* (2018), 2822–2832.

[20] Jie Huang, Wengang Zhou, Qilin Zhang, Houqiang Li, and Weiping Li. 2018. Video-based Sign Language Recognition without Temporal Segmentation. In *AAAI Conference on Artificial Intelligence*. 2257–2264.

[21] Wencan Huang, Wenwen Pan, Zhou Zhao, and Qi Tian. 2021. Towards Fast and High-Quality Sign Language Production. In *ACM International Conference on Multimedia*. 3172–3181.

[22] Ashraful Islam, Chengjiang Long, Arslan Basharat, and Anthony Hoogs. 2020. DOA-GAN: Dual-Order Attentive Generative Adversarial Network for Image Copy-move Forgery Detection and Localization. In *IEEE Conference on Computer Vision and Pattern Recognition*. 4676–4685.

[23] Ashraful Islam, Chengjiang Long, and Richard Radke. 2021. A Hybrid Attention Mechanism for Weakly-Supervised Temporal Action Localization. In *AAAI Conference on Artificial Intelligence*. 1637–1645.

[24] Songyao Jiang, Bin Sun, Lichen Wang, Yue Bai, Kunpeng Li, and Yun Fu. 2021. Skeleton Aware Multi-modal Sign Language Recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*. 3413–3423.

[25] Jestin Joy, Kannan Balakrishnan, and Sreeraj Madhavankutty. 2020. Developing a Bilingual Mobile Dictionary for Indian Sign Language and Gathering Users Experience with SignDict. *Assistive Technology* (2020), 153–160.

[26] Jichao Kan, Kun Hu, Markus Hagenbuchner, Ah Chung Tsoi, Mohammed Bennamoun, and Zhiyong Wang. 2022. Sign Language Translation with Hierarchical Spatio-temporal Graph Neural Network. In *IEEE Winter Conference on Applications of Computer Vision*. 3367–3376.

[27] Diederik P Kingma and Jimmy Ba. 2015. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations*. 1–15.

[28] Thomas N Kipf and Max Welling. 2016. Semi-Supervised Classification with Graph Convolutional Networks. *arXiv e-prints* (2016), arXiv–1609.

[29] Oscar Koller. 2020. Quantitative Survey of the State of the Art in Sign Language Recognition. *ArXiv Preprint ArXiv:2008.09918* (2020), ArXiv–2008.

[30] Taegyeong Lee, Soyeong Kwon, and Taehwan Kim. 2024. Grid Diffusion Models for Text-to-Video Generation. In *IEEE Conference on Computer Vision and Pattern Recognition*. 8734–8743.

[31] Dongxu Li, Cristian Rodriguez, Xin Yu, and Hongdong Li. 2020. Word-Level Deep Sign Language Recognition from Video: A New Large-scale Dataset and Methods Comparison. In *IEEE Winter Conference on Applications of Computer Vision*. 1459–1469.

[32] Chin-Yew Lin. 2004. Rouge: A Package for Automatic Evaluation of Summaries. In *Text Summarization Branches Out*. 74–81.

[33] Ziyu Liu, Hongwen Zhang, Zhenghao Chen, Zhiyong Wang, and Wanli Ouyang. 2020. Disentangling and Unifying Graph Convolutions for Skeleton-Based Action Recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*. 143–152.

[34] Chengjiang Long and Gang Hua. 2015. Multi-Class Multi-Annotator Active Learning with Robust Gaussian Process for Visual Recognition. In *International Conference on Computer Vision*. 2839–2847.

[35] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a Method for Automatic Evaluation of Machine Translation. In *Association for Computational Linguistics*. 311–318.

[36] Maria Parelli, Katerina Papadimitriou, Gerasimos Potamianos, Georgios Pavlakos, and Petros Maragos. 2022. Spatio-temporal Graph Convolutional Networks for Continuous Sign Language Recognition. In *IEEE International Conference on Acoustics, Speech and Signal Processing*. 8457–8461.

[37] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *IEEE Conference on Computer Vision and Pattern Recognition*. 10684–10695.

[38] Hirohiko Sagawa and Masaru Takeuchi. 2002. A Teaching System of Japanese Sign Language Using Sign Language Recognition and Generation. In *ACM International Conference on Multimedia*. 137–145.

[39] Ben Saunders, Necati Cihan Camgoz, and Richard Bowden. 2020. Progressive Transformers for End-to-End Sign Language Production. In *European Conference on Computer Vision*. 687–705.

[40] Ben Saunders, Necati Cihan Camgoz, and Richard Bowden. 2021. Mixed Signals: Sign Language Production Via A Mixture of Motion Primitives. In *International Conference on Computer Vision*. 1919–1929.

[41] Liushuai Shi, Le Wang, Chengjiang Long, Sanping Zhou, Mo Zhou, Zhenxing Niu, and Gang Hua. 2021. SGCN: Sparse Graph Convolution Network for Pedestrian Trajectory Prediction. In *IEEE Conference on Computer Vision and Pattern Recognition*. 8994–9003.

[42] Lei Shi, Yifan Zhang, Jian Cheng, and Hanqing Lu. 2019. Two-Stream Adaptive Graph Convolutional Networks for Skeleton-Based Action Recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*. 12026–12035.

[43] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. 2015. Deep Unsupervised Learning Using Nonequilibrium Thermodynamics. In *International Conference on Machine Learning*. 2256–2265.

[44] Shengeng Tang, Dan Guo, Richang Hong, and Meng Wang. 2021. Graph-Based Multimodal Sequential Embedding for Sign Language Translation. *IEEE Transactions on Multimedia* (2021), 4433–4445.

[45] Shengeng Tang, Jiayi He, Lechao Cheng, Jingjing Wu, Dan Guo, and Richang Hong. 2025. Discrete to Continuous: Generating Smooth Transition Poses from Sign Language Observations. In *IEEE Conference on Computer Vision and Pattern Recognition*. 3481–3491.

[46] Shengeng Tang, Jiayi He, Dan Guo, Yanyan Wei, Feng Li, and Richang Hong. 2025. Sign-IDD: Iconicity Disentangled Diffusion for Sign Language Production. In *AAAI Conference on Artificial Intelligence*. 7266–7274.

[47] Shengeng Tang, Feng Xue, Jingjing Wu, Shuo Wang, and Richang Hong. 2025. Gloss-Driven Conditional Diffusion Models for Sign Language Production. *ACM Transactions on Multimedia Computing, Communications and Applications* (2025), 1–17.

[48] Yansong Tang, Yi Tian, Jiwen Lu, Peiyang Li, and Jie Zhou. 2018. Deep Progressive Reinforcement Learning for Skeleton-based Action Recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*. 5323–5332.

[49] Anirudh Tunga, Sai Vidyaranya Nuthalapati, and Juan Wachs. 2021. Pose-Based Sign Language Recognition using GCN and BERT. In *IEEE Winter Conference on Applications of Computer Vision*. 31–40.

[50] Arash Vahdat, Francis Williams, Zan Gojcic, Or Litany, Sanja Fidler, Karsten Kreis, et al. 2022. Lion: Latent Point Diffusion Models for 3D Shape Generation. *Neural Information Processing Systems* (2022), 10021–10039.

[51] Manuel Vázquez-Enríquez, Jose L Alba-Castro, Laura Docío-Fernández, and Eduardo Rodríguez-Banga. 2021. Isolated Sign Language Recognition with Multi-scale Spatial-temporal Graph Convolutional Networks. In *IEEE Conference on Computer Vision and Pattern Recognition*. 3462–3471.

[52] Yongxin Wang, Kris Kitani, and Xinshuo Weng. 2021. Joint Object Detection and Multi-Object Tracking with Graph Neural Networks. In *International Conference on Robotics and Automation*. 13708–13715.

[53] ZHAO-QI Wang. 2002. A Method to Synthesize Chinese Sign Language Based on Virtual Human Technologics. *Journal of Software* (2002), 2051–2056.

[54] Pan Xie, Qipeng Zhang, Peng Taiying, Hao Tang, Yao Du, and Zexian Li. 2024. G2P-DDM: Generating Sign Pose Sequence from Gloss Sequence with Discrete Diffusion Model. In *AAAI Conference on Artificial Intelligence*. 6234–6242.

[55] Sijie Yan, Yuanjun Xiong, and Dahua Lin. 2018. Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition. In *AAAI Conference on Artificial Intelligence*.

[56] Shuai Yang, Yifan Zhou, Ziwei Liu, and Chen Change Loy. 2024. FRESCO: Spatial-Temporal Correspondence for Zero-Shot Video Translation. In *IEEE Conference on Computer Vision and Pattern Recognition*. 8703–8712.

[57] Fanfan Ye, Shiliang Pu, Qiaoyong Zhong, Chao Li, Di Xie, and Huiming Tang. 2020. Dynamic GCN: Context-Enriched Topology Learning for Skeleton-Based Action Recognition. In *ACM International Conference on Multimedia*. 55–63.

[58] Chunpeng Yu, Jiajia Liang, Yihui Liao, Zhifeng Xie, and Bin Sheng. 2022. Graph Adversarial Network with Bottleneck Adapter Tuning for Sign Language Production. In *Computer Graphics International Conference*. 92–103.

[59] Ni Zeng, Yiqiang Chen, Yang Gu, Dongdong Liu, and Yunbing Xing. 2020. Highly Fluent Sign Language Synthesis Based on Variable Motion Frame Interpolation. In *IEEE International Conference on Systems, Man, and Cybernetics*. 1772–1777.

[60] Runhao Zeng, Wenbing Huang, Mingkui Tan, Yu Rong, Peilin Zhao, Junzhou Huang, and Chuang Gan. 2019. Graph Convolutional Networks for Temporal Action Localization. In *International Conference on Computer Vision*. 7094–7103.

[61] Jihai Zhang, Wengang Zhou, Chao Xie, Junfu Pu, and Houqiang Li. 2016. Chinese Sign Language Recognition with Adaptive HMM. In *IEEE International Conference on Multimedia and Expo*. IEEE, 1–6.

[62] Mingyuan Zhang, Zhongang Cai, Liang Pan, Fangzhou Hong, Xinying Guo, Lei Yang, and Ziwei Liu. 2024. Motiondiffuse: Text-Driven Human Motion Generation with Diffusion Model. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024), 4115–4128.

[63] Rui Zhao, Liang Zhang, Biao Fu, Cong Hu, Jinsong Su, and Yidong Chen. 2024. Conditional Variational Autoencoder for Sign Language Translation with Cross-modal Alignment. In *AAAI Conference on Artificial Intelligence*. 19643–19651.

[64] Weichao Zhao, Hezhen Hu, Wengang Zhou, Yunyao Mao, Min Wang, and Houqiang Li. 2024. Masa: Motion-Aware Masked Autoencoder with Semantic Alignment for Sign Language Recognition. *IEEE Transactions on Circuits and Systems for Video Technology* (2024).

[65] Weichao Zhao, Hezhen Hu, Wengang Zhou, Jiaxin Shi, and Houqiang Li. 2023. BEST: BERT Pre-Training for Sign Language Recognition with Coupling Tokenization. In *AAAI Conference on Artificial Intelligence*. 3597–3605.

[66] Wenyang Zhou, Zhiyang Dou, Zeyu Cao, Zhouyingcheng Liao, Jingbo Wang, Wenjia Wang, Yuan Liu, Taku Komura, Wenping Wang, and Lingjie Liu. 2024. Emdm: Efficient Motion Diffusion Model for Fast and High-Quality Motion Generation. In *European Conference on Computer Vision*. 18–38.

[67] Ronglai Zuo, Fangyun Wei, and Brian Mak. 2023. Natural Language-Assisted Sign Language Recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*. 14890–14900.